Subsea Lens - Customer Brief

Subsea pipelines carry oil and gas from offshore fields. Operators monitor pressure along these pipelines to spot possible problems and decide when to intervene. A missed leak can harm the marine environment, stop production and put operating permissions at risk. But sending a vessel or underwater inspection team to investigate faulty measurements can also be costly. The integrity engineer must therefore distinguish a real pipeline problem from a measurement problem before directing a response.

Making that distinction is difficult because different causes can produce similar alerts. A pressure drop may indicate a leak, an operating change or a faulty sensor; missing readings can obscure the picture. A leak and sensor faults may also occur together. Before directing a response, the engineer must establish what is happening, where attention is needed and whether the evidence is reliable enough to act on. This turns each investigation into three practical questions:

1. Is this a leak, a change in operating or measurement conditions, or an instrument problem?

2. If a leak is suspected, which branch and section should be inspected?

3. What evidence supports that conclusion, and what still needs checking before a team is sent?

The illustration below shows this challenge on a simulated 11.2 km network with seven connected sections (P1-P7) and 112 pressure instruments. A leak on P5 occurs alongside sensor faults, missing data and a shared data-connection problem on P7. Both maps summarize the same 60-second slice of simulated readings of scenario S7 (see the live tab of the dashboard), from 50 to 110 seconds into the replay, with 69 instruments flagged.

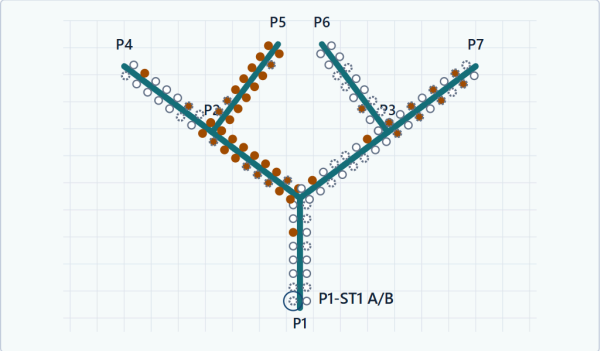


Figure 1. Sensor alerts only. Orange dots show unusual pressure; dashed dots show missing or invalid readings. Question 1 requires comparing histories and records to separate the P5 leak from P7's connection problem and sensor faults. Question 2 requires tracing related pressure changes across branches to narrow the inspection area. Question 3 requires checking sensor reliability and documenting the supporting evidence. The map alone answers none of these: repeated cross-checks take time and require knowledge of pipeline behavior.

Subsea Lens organizes these cross-checks with **three AI agents working in sequence**. Sensor Quality checks reading reliability. Event Diagnosis uses that assessment, pressure patterns and records to assess causes and inspection priorities. Operator Response assembles the findings, evidence, uncertainties and next checks into one report. A separate review checks the report against its sources.

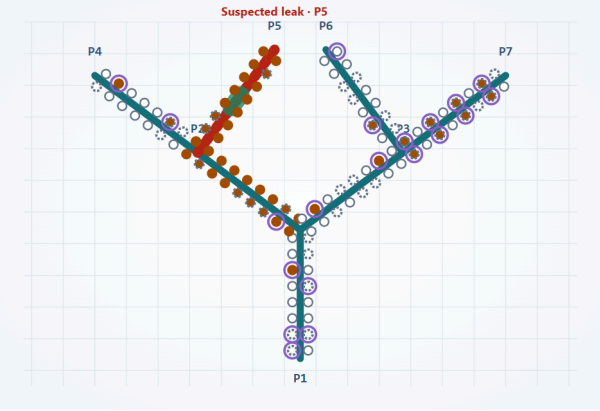


Figure 2. Applying this workflow to the same readings. For question 1, the report separates a suspected P5 leak (red) from sensor/data concerns (purple rings, not confirmed hardware faults). For question 2, green marks the report's inspection priority, 700-900 m from P5's start, not an exact leak point. For question 3, the report links its conclusions to readings and records and states what still needs checking. The engineer reviews a connected explanation rather than assembling it from individual alerts, and retains the decision on inspection and deployment.

This handoff also needs to be timely. In two controlled tests of this simulated scenario, reviewed reports were ready in about 2.5-2.8 minutes of elapsed time.

To test the system's answers, we compared its findings with known leaks and instrument problems that were hidden from the agents. Across all evaluated snapshots in the held-out synthetic detection test, leak detection achieved **100% precision and recall, with no false leak alarms**. Identifying affected instruments and classifying their problems also achieved **100% precision and recall**.

Run the demo. Dashboard evidence. Evaluation and timing results.